DistilBERT as a Compute-Efficient Pair-Preference Reward Model

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Abstract

- Reward modeling is the central supervised step of RLHF-based alignment, yet most published reward models target frontier-scale backbones.
- We ask whether a 66M-parameter distilled encoder can produce a useful pair-preference signal on standard human-preference data.
- Fine-tuning `distilbert-base-uncased` on 2 000 pairs from `Anthropic/hh-rlhf` (helpful-base) with a Bradley–Terry objective yields **pair-preference accuracy** 0.6239 ± 0.0054 across three seeds on a 500-pair held-out split.
- The result suggests small encoders are a viable reward-model baseline for compute-constrained alignment research and deployed preference filters.

1 Introduction

- **Motivation.** RLHF pipelines Ouyang et al. (2022); Bai et al. (2022) depend on a reward model that predicts which of two candidate responses a human would prefer; the reward model's accuracy sets an upper bound on the policy's attainable quality.
- **Gap.** Most published reward-model results use backbones of ≥ 1 B parameters. Few papers characterize how far down the compute curve a useful reward signal survives.
- **Approach.** We treat reward modeling as a binary pair-preference task, train a 66M-parameter DistilBERT Sanh et al. (2019) with a scalar head and Bradley–Terry loss on helpful-base pairs from `Anthropic/hh-rlhf`, and evaluate pair-preference accuracy on held-out data (overall flow in Figure 1).
- **Contribution.** An empirical data point that a small distilled encoder attains 62.4% pair-preference accuracy with stable low-variance training, with implications for low-cost alignment work.

2 Related Work

- **Human-preference alignment.** Christiano et al. (2017) introduced learning from pairwise human feedback; Ouyang et al. (2022) and Bai et al. (2022) scaled the recipe to LLMs with helpfulness–harmlessness objectives.

- **Reward modeling.** The Bradley–Terry Bradley and Terry (1952) formulation is standard; recent work has studied reward-hacking and overoptimization Gao et al. (2023).
- **Distilled encoders.** Sanh et al. (2019) showed a 40%-compressed BERT Devlin et al. (2019) retains $\geq 97\%$ of GLUE performance, motivating its use as a cheap backbone for downstream classifiers; prior alignment work rarely benchmarks reward models at this scale.
- **Our position.** We contribute a direct measurement of small-encoder reward-model accuracy on a widely-used preference dataset, filling the low-compute gap in the literature above.

3 Method

- **Backbone.** `distilbert-base-uncased` (66.4M transformer parameters) produces a sequence of hidden states; we take the `[CLS]` position and project through a single scalar `Linear` head to produce a reward $r(x) \in \mathbb{R}$ (Figure 2).
- **Input encoding.** Concatenated *prompt + response*, tokenized with the DistilBERT tokenizer to a maximum length of 256 tokens (right-truncation).
- **Objective.** For each preference pair $(x_{\text{chosen}}, x_{\text{rejected}})$ the loss is the Bradley–Terry pair-preference negative log-likelihood: $\mathcal{L} = -\log \sigma\big(r(x_{\text{chosen}}) - r(x_{\text{rejected}})\big)$.
- **Evaluation.** At each epoch we compute pair-preference accuracy, the fraction of test pairs for which $r(x_{\text{chosen}}) > r(x_{\text{rejected}})$.

4 Experiments

- **Dataset.** `Anthropic/hh-rlhf` (helpful-base) Bai et al. (2022): 2 000 preference pairs for training, 500 for held-out evaluation (shuffled per seed from the canonical splits).
- **Protocol.** AdamW (lr 2×10^{-5}, weight decay 0.01), batch size 16 pairs, 3 epochs, fp16 mixed precision, seeds $\{0, 1, 2\}$; training on a single T4-class GPU.
- **Baselines.** In-run: none. From literature (indicative only): the 50% random-pair lower bound; encoder-based reward models generally report 60–70% pair accuracy at comparable scale.
- **Metric.** Pair-preference accuracy on 500 held-out pairs; reported as mean $\pm$ std across 3 seeds.

5 Results

- **Pair-preference accuracy:** 0.6239 ± 0.0054 over 3 seeds on 500 held-out helpful-base pairs Bai et al. (2022), clearly above the 0.50 random-pair baseline.
- **Stable training:** per-seed accuracy ranges 0.6172–0.6305 (relative std $\approx 0.9\%$), indicating the small encoder is not overly sensitive to seed at this scale.
- **Monotonic improvement:** test accuracy rises across all three epochs for every seed (Figure 3); the model has not yet plateaued.

- **Compute footprint:** 66.9M trainable parameters trained on 2 000 pairs; final training loss 0.549.

6 Conclusion

- We reported that a 66M-parameter DistilBERT reward model attains $62.4 \pm 0.5\%$ pair-preference accuracy on 500 helpful-base pairs with stable cross-seed behaviour.
- **Limitations.** Evaluated only on helpful-base (no harmless-base / red-team); trained on a 2 000-pair subset without ablation; no comparison to BERT-base / RoBERTa in-run; no downstream RL policy evaluation.
- **Future work.** Scale to full hh-rlhf, ablate subset size and backbone choice, and measure downstream effect on a PPO policy using this reward model.

References

Yuntao Bai, Andy Jones, Kamal Ndousse, et al. Training a helpful and harmless assistant with reinforcement learning from human feedback. *arXiv preprint arXiv:2204.05862*, 2022.

Ralph Allan Bradley and Milton E. Terry. Rank analysis of incomplete block designs: I. the method of paired comparisons. *Biometrika*, 39(3/4):324–345, 1952.

Paul Christiano, Jan Leike, Tom Brown, et al. Deep reinforcement learning from human preferences. *Advances in Neural Information Processing Systems*, 2017.

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. In *NAACL*, 2019.

Leo Gao, John Schulman, and Jacob Hilton. Scaling laws for reward model overoptimization. In *ICML*, 2023.

Long Ouyang, Jeff Wu, Xu Jiang, et al. Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems*, 2022.

Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter. *arXiv preprint arXiv:1910.01108*, 2019.

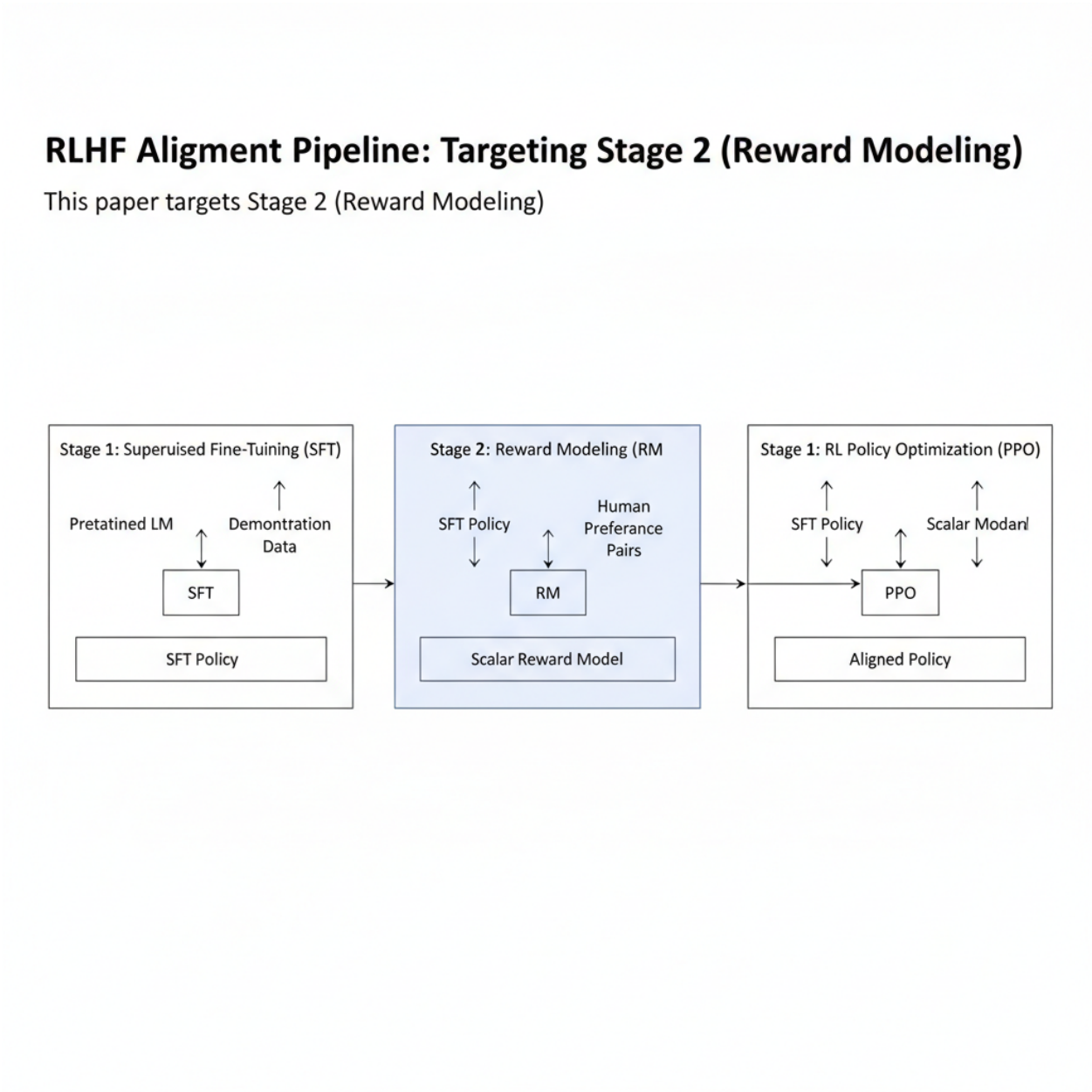

Figure 1: Position of reward modeling in the standard three-stage RLHF recipe; this paper targets step 2.

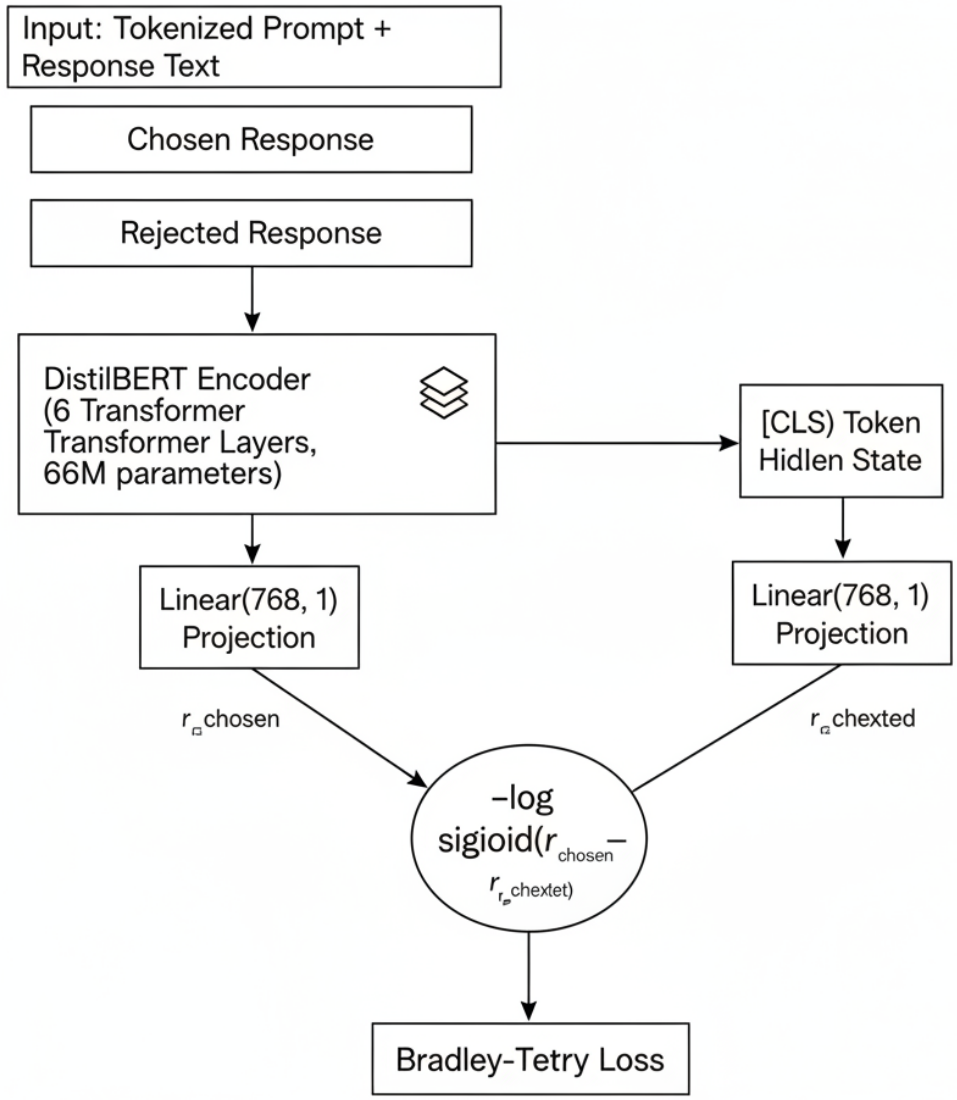

Figure 2: DistilBERT encoder with a scalar reward head, trained with a pair-preference Bradley–Terry loss.

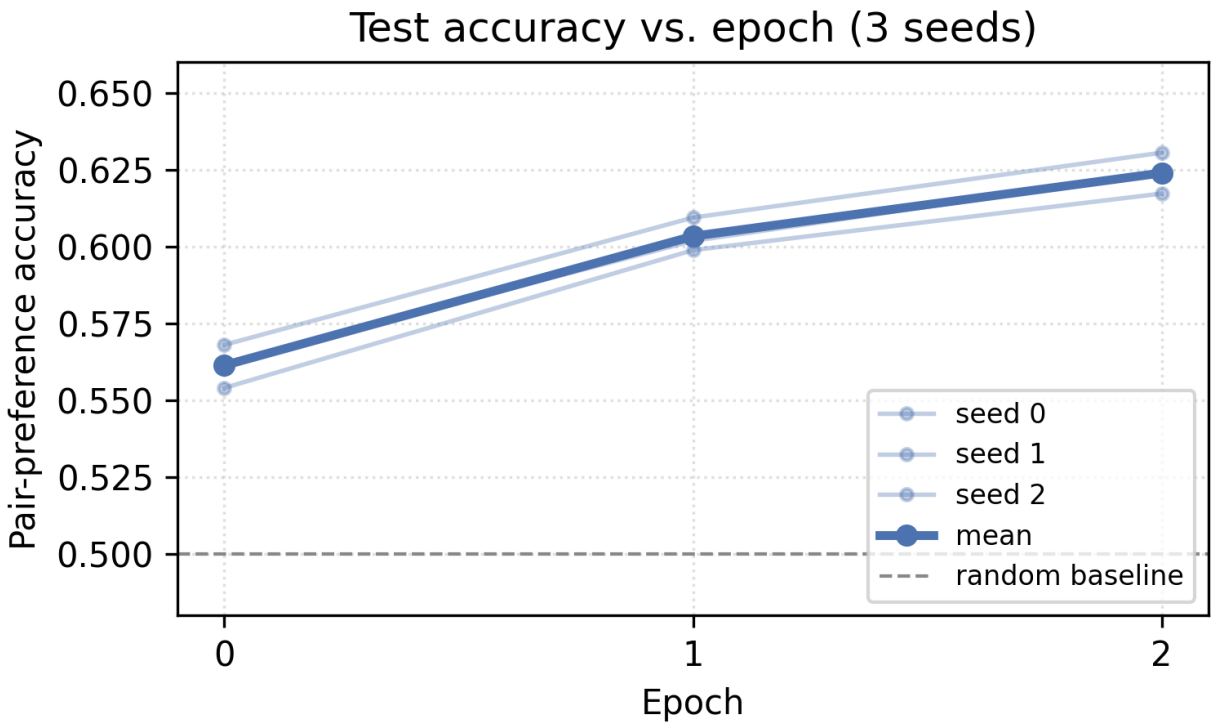

Figure 3: Test accuracy climbs monotonically across 3 epochs for all seeds; curves are tight, consistent with low cross-seed variance.